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THE UNIVERSITY OF HONG KONG

COMP7705

MSc(CompSc) Project

Project Progress Report

An innovative AI-powered interview assessment tool for company

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1. Abstract

The global AI recruitment market is growing rapidly, with companies like Amazon using AI for 60-70% of initial applicant interactions. "Candidates who are identified with AI job matching technology and reach the interview stage are 24% more likely to receive a positive outcome after their initial interview loop" (**Parulekar, 2026**). In Hong Kong, AI adoption in HR is accelerating amid talent shortage.

In this study, we aim to develop an innovative AI-powered virtual interviewer tailored for HR teams and hiring managers to streamline initial candidate screening for technical and behavioral roles.

We will use Python, web interface, and AI API as the core LLM engine. The outcome will demonstrate benefits (efficiency, consistency) and address challenges (bias, privacy) in AI recruitment.

2.Objective

The primary objectives are to:

- Develop a basic AI tool that automates resume screening by matching candidate qualifications to job descriptions, reducing human effort
- Generate customized interview questions based on job types
- Analyze candidate responses for relevance and quality, providing a ranking score to shortlist top applicants
- Demonstrate how AI can minimize bias and improve efficiency in hiring, while learning practical skills in Python, web apps, and data processing.

3.Milestones

Tasks	Estimated completion time	Estimated number of learning hours
~~1. Research & Planning~~ ~~- Review AI Hiring PDF and related literature~~ ~~- Define objectives, scope, and stream (Multimedia Computing)~~ ~~- Install Python, Streamlit, OpenAI/Groq API - Draft and submit Detailed Project Proposal~~ ~~- Create and initialize project webpage on school account~~	~~March 10, 2026~~	~~60 hours~~
~~2. Basic Interface & Question Generation~~ ~~- Build simple chat interface using Streamlit~~ ~~- Implement basic question generation with LLM API (prompt engineering)~~ ~~- Test with sample job descriptions~~ ~~- Update project webpage with screenshots and progress~~	~~April 7, 2026~~	~~50 hours~~
3. Multi-turn Conversation & Initial Scoring - Add conversation history management for multi-turn interviews - Develop initial automated scoring (relevance, technical depth) - Conduct preliminary tests with	May 5, 2026 (Progress Update 2)	45 hours

dummy responses - Update webpage with demo video/screenshots		
4. Advanced Features & Interim Report - Implement full scoring dimensions (clarity, value alignment) - Add intelligent follow-up probes - Generate basic reports and recommendations - Prepare and deliver Interim Report & Presentation	June 1, 2026 (Interim Report & Presentation)	50 hours
5. Testing, Refinement & Bias Check - Full system end-to-end testing with 20–30 dummy cases - Check for bias and improve prompts - Fix bugs and optimize performance Update webpage with test results	June 16, 2026 (Progress Update 3)	70 hours
6. Final Polish & Oral Preparation - Refine UI and report format - Conduct final full tests and collect results - Prepare demo video and oral examination materials - Final webpage update	July 6, 2026 (Progress Update 4)	50 hours

7. Final Submission - Complete Project Report - Submit final webpage (July 13)	July 17, 2026 (Project Report)	30 hours

4.Current Implementation Progress: Task 1 & 2 Completed

Phase 1 and Phase 2 are fully completed. The system is a production-ready text-based AI interview tool with the following features:

- Admin Panel: This is the core of AI-powered interview assessment tool. The company can configure company name, core values, total number of questions, maximum follow-up questions, fixed questions set by HR and AI-generated questions. **As shown in Figure 1**, we demonstrate the user-friendly setup page where all interview parameters can be easily customised.
- AI Question Generation: Uses Grok-4-1-fast-reasoning to dynamically create additional questions based on company values while avoiding duplication with fixed questions.
- Intelligent Multi-turn Conversation: The AI interviewer start with greeting by AI interviewer and entering the name **as shown in Figure 2.** Then, AI interviewer asks behavioural questions based on company values and the specific question which the company has set already while the candidate's responses in the conversation. The system automatically decides whether to ask follow-up questions or move to the next main question according to the answer. The interview will end by the default number of questions. Finally ,the

whole conversation **as shown in Figure 3** will be recorded for evaluate and further usage.

- English Enforcement: Custom language detection function ensures candidates respond in English.
- SQLite Database: Persistent storage of all company settings and interview records (conversation + AI report + score).
- Automatic Report Generation: After completion, the system produces a professional evaluation report including overall score (0–100), strengths, weaknesses, value alignment and hiring recommendation. **As shown in Figure 4**, this is an example of the comprehensive AI-generated evaluation report produced after an interview. It includes overall score, performance analysis for each question and follow-up, candidate strengths, areas for improvement, alignment with company values, communication skills assessment and final hiring recommendation.
- Report History & Ranking: In the Admin Panel's "Interview Reports" tab, the system displays a ranked list of all candidates who have completed the interview. Each entry shows the candidate's name, overall score (0-100) and timestamp logs **as shown in Figure 5**. This screenshot highlights the ranking feature and the expandable detailed report section and full conversation.

The entire system is built as a single-file Streamlit application using **Python which is attached in the report**, xAI Grok API, and SQLite — exactly matching the technology stack described in the Detailed Project Proposal.

5.System Flow Overview

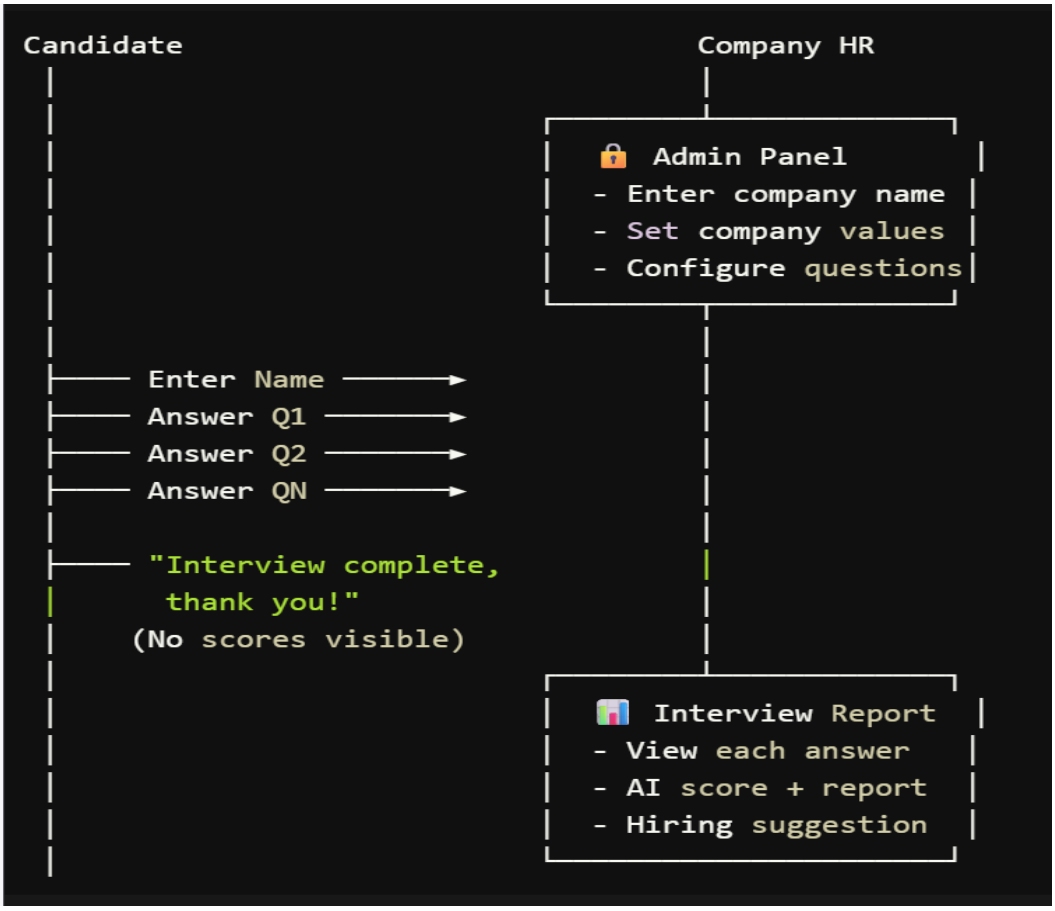
The system consists of two main sides: the Candidate side and the

Company HR side.

On the HR side, administrators can securely access the backend to configure the company name, core values, and interview questions.

Candidates begin by entering their full name and then answer a series of questions sequentially. After completing all questions, they receive a simple "Interview Completed. Thank you!" message and cannot view any scores or reports.

On the HR side, the system generates a comprehensive Interview Report for each candidate. This report includes the full conversation history, AI-generated scores, detailed performance analysis, and hiring recommendations.



6.Future Work (Task 3 to 7)

The foundation is now solid. The next steps follow the plan to develop the AI-powered interview assessment tool :

- Add voice conversation (speech-to-text + text-to-speech).
- Add virtual interviewer avatar.
- Implement camera system with anti-deepfake protection (liveness, deepfake detection, continuous identity verification).
- Facial expression analysis (smile, tension, eye contact).
- Voice tone analysis (speed, pitch, confidence).
- Final comprehensive report combining text score + expression + tone + authenticity.

7.Challenges

Several technical and planning challenges have been encountered when building multimodal AI systems.

- One major upcoming challenge is implementing a virtual interviewer avatar which is the technique point and selling point to more reality of the tool. We need to research suitable tools to create a realistic, animated avatar with lip-sync capability. We must balance realism, performance, and integration cost with Streamlit.
- Another concern is speech-to-text functionality. Although many tools are available, we are worried about real-world quality. We will need extensive testing to ensure high accuracy in noisy environments and different accents, as poor speech recognition could significantly affect user experience.
- Finally, latency remains a critical issue. The current Grok API already introduces a small delay during multi-turn conversation.

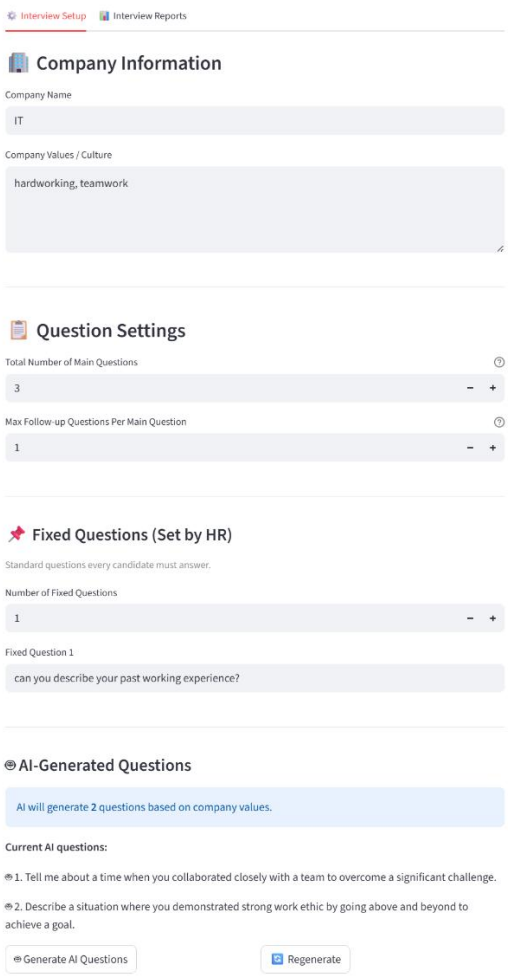
When we add voice input, avatar animation, facial analysis, and tone detection in later phases, the cumulative delay may become noticeable. We are researching optimization techniques to switch to faster inference models to maintain real-time responsiveness.

8.Project Website

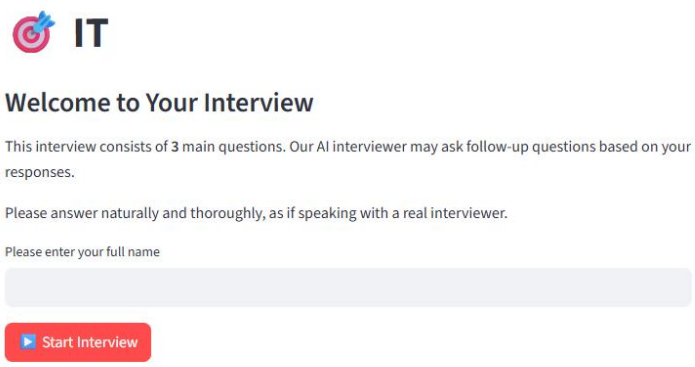
The project webpage is live at https://kyssai.github.io/FYP/FYP/index.html and has been regularly updated with screenshots, progress updates, and live demo links.

9.Figures

1)



2)


IT
Welcome to Your Interview
This interview consists of 3 main questions. Our AI interviewer may ask follow-up questions based on your responses.
Please answer naturally and thoroughly, as if speaking with a real interviewer.
Please enter your full name
Start Interview

3)

Full Conversation

Interviewer: Hello chan tai man! Welcome to the interview at software engineer. I'll be your interviewer today. Let's get started.

what is your advantage?

chan tai man: harworking

Interviewer: That's a great strength, especially here at our company. Can you share a specific example from your experience where your hard work made a difference in a project or team?

chan tai man: left company late 2 hours

Interviewer: That's dedication—staying late shows commitment! What was the project you were working on, and how did those extra hours contribute to its success?

chan tai man: no project just routine work

Interviewer: Thanks for sharing that—staying committed to routine work is a solid foundation for success. To help me understand your background better, can you describe your past working experience?

chan tai man: Created a new format for presenting and reporting sales, customer engagement and Google Ad reports that shortened the number of meetings by about 30% during the last three quarters. Monitored and updated the bid strategy for Google Ads, which resulted in a Click-Through Rate (CTR) increase of 4.5% in the first three months. Redesigned and conducted keyword research for product pages of the online store, which increased the organic keywords in Top 100 by 5,600 and in Top 10 by 315 for high-volume searches (over 5,000 monthly clicks). Located and suggested new potential business partnerships by reaching out to potential partners and taking part in professional networking events which resulted in three new strategic partnerships.

Interviewer: That's impressive work—those quantifiable improvements in efficiency, performance, and partnerships really stand out and align well with our focus on innovation and results. Moving on, can you describe a time when you introduced an innovative solution to a technical challenge in a software project? What was the problem, your approach, and the results?

chan tai man: can you repeat

Interviewer: Sure thing! Can you describe a time when you introduced an innovative solution to a technical challenge in a software project? What was the problem, your approach, and the results?

chan tai man: Anyone can propose wild ideas; valuable innovation balances creativity with feasibility. Interviewers assess whether you can implement creative ideas practically, work within real-world constraints creatively, pilot and refine innovations iteratively, and measure innovation impact objectively.

Interviewer: That's an insightful take on what makes innovation effective—balancing creativity with practicality is key. To dive deeper, can you share a specific example from a software project where you introduced such a solution? What was the challenge, how did you approach it, and what were the outcomes?

chan tai man: no challenge encounter before

Interviewer: Thanks for your honesty—it's okay if you haven't faced that specific scenario yet; experience comes in many forms. Shifting gears to teamwork, which is huge here at our company: Tell me about a collaborative software development project where teamwork was essential. What was your role, how did you contribute to the team's success, and what did you learn?

chan tai man: etailed Context: It clearly defines the challenge and your team. Focused Contribution: Highlights your specific role and actions taken. Measurable Result: The mention of sales exceeding expectations by 20% provides a tangible result. Team Collaboration Emphasized: This answer showcases how your leadership and coordination skills directly contributed to team success.

4)

AI Evaluation Report

1. Overall Score (0-100)

10/100

2. Performance analysis for key questions and follow-ups

- **Past working experience**: Candidate responded with "IT", which is vague and uninformative. No details on duration, role, or achievements were provided.
- **Elaboration on IT role**: Response was "coding", again extremely brief. Failed to specify responsibilities, technologies, or impact.
- **Teamwork example (core company value)**: Initial response "no sorry" directly refused to answer. Even after probing for a small example, candidate asked to repeat the question ("please repeat your question"), showing disengagement rather than effort.
- **Work ethic example (core company value)**: Responded "sorry i dun know", indicating lack of preparation or relevant experience. Follow-up prompt for a smaller example yielded "testing", which appears irrelevant or nonsensical. Overall, responses lacked substance, and the interviewer had to repeatedly rephrase or simplify questions, with no meaningful engagement from the candidate.

3. Candidate's strengths

- Showed basic politeness (e.g., using "sorry" in responses).
- Participated minimally by responding to every question, rather than going silent.
- No major red flags like rudeness or unprofessional behavior.

4. Areas for improvement

- **Preparation**: Candidate seemed unprepared with no specific examples, stories, or details from past experience ready to share. Behavioral questions require STAR (Situation, Task, Action, Result) structured responses.
- **Depth and relevance**: Answers were one-word or dismissive; needs to expand on experiences with concrete examples.
- **Engagement**: Repeatedly asked for question repetition or gave non-answers, suggesting lack of focus or interest.
- **Professionalism**: Informal language like "dun know" (instead of "don't know") detracts from credibility.

5. Alignment with company values

- **Teamwork**: Poor alignment—no examples provided despite multiple prompts. Directly stated "no sorry," indicating no relevant experience or unwillingness to share.
- **Hardworking**: No demonstration of work ethic; responses like "sorry i dun know" and irrelevant "testing" fail to show going "above and beyond." Overall, zero evidence of fit with IT's hardworking, teamwork-oriented culture.

6. Communication skills assessment (clarity, confidence, depth of answers)

- **Clarity**: Low—responses were terse, vague (e.g., "IT", "coding"), and sometimes off-topic ("testing").
- **Confidence**: Very low—evident in apologies, refusals ("no sorry"), and requests to repeat questions, suggesting uncertainty or discomfort.
- **Depth of answers**: Non-existent; no elaboration, metrics, or context provided. Answers lacked structure and professionalism.

7. How well they handled follow-up questions

Poorly. The interviewer provided multiple gentle probes and simplifications (e.g., "small example from your coding work"), but candidate either repeated requests (e.g., "please repeat your question"), apologized without answering, or gave irrelevant responses. This showed inability to adapt, pivot, or build on prompts, turning a supportive interview into a one-sided effort.

8. Final hiring recommendation

Do Not Recommend

5)

Admin Panel

Log Out

Interview Setup | Interview Reports

Candidate Interview Reports

5 candidates have completed the interview

Candidate Ranking

#1 Figo — **65** / 100 (2026-04-14 03:06:51)

#2 chan tai man — **35** / 100 (2026-04-14 02:53:43)

#3 Peter — **15** / 100 (2026-04-14 03:33:04)

#4 testing — **10** / 100 (2026-04-14 05:27:26)

#5 David — **10** / 100 (2026-04-14 04:32:29)

Detailed Reports

- **testing** — 10 / 100 (2026-04-14 05:27:26)
- **David** — 10 / 100 (2026-04-14 04:32:29)
- **Peter** — 15 / 100 (2026-04-14 03:33:04)
- **Figo** — 65 / 100 (2026-04-14 03:06:51)
- **chan tai man** — 35 / 100 (2026-04-14 02:53:43)